

WEALTH vs HEALTH

The Global Life Expectancy Paradox

Does money buy a longer life? Correlating GDP, schooling, and immunization with life expectancy across nations.

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An Age-Old Question, Answered With Data

Every country wants the same thing for its people: a long, healthy life. But the paths nations take to get there look very different. Some invest heavily in economic growth; others prioritize education and public health. This project uses WHO life-expectancy data to test a simple but important idea — that money alone does not determine how long people live.

190+**Countries Covered****20+****Years of Records****3****Core Predictors Studied**

The Global Life Expectancy Paradox

Wealthier nations are generally assumed to guarantee healthier, longer-living populations. Yet life expectancy varies sharply even among countries with similar income levels — revealing that health outcomes depend on more than GDP alone.

1

Correlate

GDP, schooling, and immunization rates against life expectancy to quantify their real influence.

2

Compare

Contrast developed vs. developing nations to expose where the “wealth = health” assumption breaks down.

3

Handle Gaps

Address missing and inconsistent records, common in data from developing nations, without biasing results.

Why This Study Matters

1

Determine which factor — wealth, education, or immunization — has the strongest statistical relationship with life expectancy.

2

Provide evidence-based insight for policymakers on where health investment yields the greatest returns.

3

Challenge the assumption that GDP growth automatically translates into a longer, healthier life for citizens.

4

Build a reusable analytical approach for handling incomplete health data from low-reporting-capacity nations.

Scope of the Study

In Scope

- ✓ Country-wise WHO life expectancy dataset, multi-year records
- ✓ GDP, schooling years, and immunization coverage as core predictors
- ✓ Comparison across developed and developing economies
- ✓ Statistical correlation and visual trend analysis

Out of Scope

- ✗ Individual-level medical or clinical patient data
- ✗ Real-time or post-dataset-cutoff health events
- ✗ Causal claims beyond statistical correlation
- ✗ Non-WHO-reported member states or territories

Global Data: GDP, Life Expectancy & More

Sourced from Kaggle (arslaan5/global-data-gdp-life-expectancy-and-more), an aggregation of WHO and UN health & economic indicators compiled per country, per year.

Status

Developed vs. Developing classification

Adult / Infant Mortality

Deaths per 1,000 population

Schooling

Average years of schooling

BMI, Alcohol, Health Expenditure

Lifestyle & spending indicators

Life Expectancy

Target variable, in years

GDP

Gross Domestic Product per capita

Immunization

Hepatitis B, Polio, Diphtheria coverage %

Population, HIV/AIDS

Demographic & epidemiological context

The Three Predictors We Focus On

GDP

WEALTH

Economic output per person — the proxy for a nation's material resources.

Schooling

EDUCATION

Average years of formal education — a proxy for knowledge & opportunity.

Immunization

PUBLIC HEALTH

Coverage of Hepatitis B, Polio & Diphtheria vaccines in children.

Target Variable: Life Expectancy (years) — measured against each predictor above.

GDP per Capita vs. Life Expectancy

Life expectancy rises steeply at low incomes, then flattens beyond roughly \$20,000 GDP per capita — a classic diminishing-returns curve. Extra wealth still helps, but each dollar buys less additional lifespan once basic needs are met.

Insight: Wealth matters most for the poorest nations — not the richest.

Schooling vs. Life Expectancy

Schooling shows one of the strongest, most linear relationships in the dataset. Each additional year of average education tracks closely with several added years of life expectancy — more consistently than GDP does.

Insight: Education may be a more reliable predictor than income alone.

Immunization Coverage vs. Life Expectancy

Immunization coverage correlates with longer life, but with more scatter than schooling — some nations achieve high vaccine coverage despite low income, showing that targeted public-health programs can outperform GDP as a lever for change.

Insight: Public-health policy can raise life expectancy even where GDP is low.

Handling Missing Data in Developing Nations

Developing nations report health statistics less consistently, leaving gaps concentrated exactly where the paradox is most interesting to study.

1

Identify

Flag missing values by country & indicator; visualize gap concentration by region and income group.

2

Impute

Fill gaps using country-group median / regression-based imputation (grouped by Status: Developed vs. Developing) instead of a single global average.

3

Validate

Cross-check imputed values against neighboring years and regional trends to avoid distorting the wealth–health relationship.

4

Flag & Report

Retain a missing-data indicator column so analysis can be re-run with and without imputed records for transparency.

CONCLUSION

Does Money Buy a Longer Life? Not Alone.

GDP

Matters most at low incomes; effect flattens sharply after basic needs are met.

Schooling

The most consistent predictor — education tracks closely with added years of life.

Immunization

Proves policy can beat poverty — high coverage lifts life expectancy even at low GDP.

Wealth opens the door, but education and public health decide how far a nation walks through it. Sustainable gains in life expectancy come from balanced investment — not GDP growth alone.